MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSCCSC201 - DATA STRUCTURES USING C

Duration of the Examination: **1.5 Hrs** Maximum Marks: **35**

List of Questions

***** All questions belong to the **Apply** level and are aligned with Course Outcome 4 (**CO4**).

1. Program using function to compute the sum of digits of a number.
2. Program using a function that returns the sum of the first N natural numbers.
3. Develop a program that uses a function to reverse a number.
4. Program using function to calculate the factorial of a given number.
5. Program using function to print the fibonacci series.
6. Program to insert an element at a specific position in an array.
7. Program to delete an element from a specific position in an array.
8. Program to find the largest and smallest element in an array.
9. Program to count even and odd numbers in an array.
10. Program to reverse the elements of an array.
11. Implement Linear Search to find an element in an unsorted array.
12. Implement Binary Search to search an element in a sorted array.
13. Implement Bubble Sort to arrange an array of integers in descending order.
14. Program to sort an array using Merge Sort.
15. Implement push and pop operations in a stack using arrays.
16. Implement queue operations using arrays.
17. Implement Circular Queue using arrays with operations: enqueue, dequeue, and display.
18. Implement creation and display of a singly linked list.
19. Program to delete an element from the first and last position from a single linked list.
20. Program to search an element in a singly linked list.